RW-6769 **421201**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Tamil

தமிழ் உரைமரபுகள்

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

எல்லா வினாக்களும் சம மதிப்பெண் உடையவை.

எல்லாக் கேள்விகளுக்கும் கட்டுரை வடிவில் விடையளிக்க வேண்டும்.

(5 × 15 = 75)

1. (a) தான் எடுத்துக்கொண்ட கருத்தை நிறுவுவதில் சேனாவரையர் வெற்றி பெறும் முறைகளை எழுதுக.

(அல்லது)

(b) நச்சினார்க்கினியர் மீறா கருத்துக்களை மறுக்கும் முறைகள் குறித்து எழுதுக.

2. (a) வைணவ உரைவளம் குறித்து எழுதுக.

(*அல்லது*)

(b) சிவஞான முனிவர் உரைகளில் காணப்படும் சமய நெறி விளக்கங்கள் குறித்து எழுதுக.

3. (a) தொல்காப்பிய மூலத்தை எளிமையாக விளக்குவதில் இளம்பூரணர் பெற்ற சிறப்பை விளக்குக.

(*அல்லது*)

(b) வர்ணனைத் திறம் மிகுந்த களவியல் உரையில் வெளிப்படும் பழந்தமிழர் வாழ்வியல் நெறிகள் குறித்து எழுதுக.

4. (a) பரிமேலழகர் உரைகளில் வெளிப்படும் பல்துறைச் சிந்தனைகள் குறித்து எழுதுக.

(*அல்லது*)

(b) பரிமேலழகர் வடமொழிக் கோட்பாட்டுடன் உரை எழுதும் முறைகளை விளக்குக.

5. (a) மூல நூலாசிரியனுக்கு முரண்படாமல் உரையாசிரியர்கள் உரை எழுதிய முறைகளைத் தருக.

(அல்லது)

(b) உரைகளில் வெளிப்படும் பொது ஆய்வு முறைகள் குறித்து எழுதுக.

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RW-6770 **761201**

M.Phil. DEGREE EXAMINATION APRIL 2011

English

AMERICAN LITERATURE

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

Each question carries 15 marks.

1. (a) Attempt an essay on poetic techniques adopted by Emily Dickinson in " Because I could not stop for Death" and Success is Counted Sweetest".

(*Or*)

(b) Attempt an estimate of E.O. Poe as a poet with reference to the prescribed poems.

2. (a) Analyse the theme of Sylvia Plath's poems " Daddy" and " Tulips".

(*Or*)

(b) Critically examine Langston Hughe's poems " Black Panther " and " Dinner guest Me"

3. (a) Estimate the greatness of Emerson as a prose writer with reference to " The American Scholar".

(*Or*)

(b) Attempt a critical estimate of Henry James " The Art of Fiction".

4. (a) Attempt a critical appreciation of Edward Albee's " The Zoo Story ".

(*Or*)

(b) Bring out the Central theme in Arthur Miller's play " *Death of a Salesman*".

5. (a) Trace the central theme of Theodere Dreiser's " Sister Carrie".

(*Or*)

(b) Discuss the biblical elements in " The Old Man and the Sea".

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RW-6771 **571203**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Mathematics

MEASURE THEORY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

1. (a) State and Lebesgue's Monotone Convergence Theorem.

(*Or*)

(b) (i) State and Prove Fatou's Lemma.

(8)

(ii) Suppose $f : X \to [0, \infty]$ is measurable, and $\phi(E) = \int_E f d\mu \, (E \in m)$. Then prove that ϕ is a measure on m, and $\int_X g d\phi = \int_X g f d\mu$ for every measurable g on X with range in $[0, \infty]$.

(7)

2. (a) Let X be a locally compact Hausdorff space, and let $\wedge$ be a positive linear functional on $C_c(X)$.

(i) If $E_1, E_2, \ldots$ are arbitrary subsets of X, then prove that $\mu\left(\left[\bigcup_{i=1}^{\infty} E_i\right]\right) \leq \sum_{i=1}^{\infty} \mu(E_i)$.

(8)

(ii) If K is Compact, then prove that $K \in m_F$.

(7)

(*Or*)

(b) Prove that there exists a positive complete measure m defined on a σ - algebra $m\, m R^k$ with the following properties.

(i) $m(W) = \text{volu}(W)$ for every K-cell W.

(7)

(ii) m contains all Borel sets in R^k; $E \in m$ if any only if there are sets A and $B \subset R^k$ such that $A \subset B \subset \overline{B}$, A is an $F\sigma$, B is a $G\delta$ and $m(B - A) = 0$. (8)

3. (a) (i) State and prove Jensen's Inequality. (9)

(ii) State and prove Holder's Inequality. (6)

(*Or*)

(b) (i) Prove the $L^p(\mu)$ is a complete metric space, for $1 \le p \le \infty$ and for every positive measure μ. (9)

(ii) Let S be the class of all complex, measurable, simple functions on X such that $\mu(\{x : s(x) \neq 0\}) < \infty$. If $1 \le p < \infty$, Then prove that S is dense in $L^p(\mu)$. (6)

4. (a) Let μ and λ be positive bounded measures on a σ-algebra m in a set X.

(i) Prove that there is a unique pair of measures λ_a and λ_s on m such that $\lambda = \lambda_a + \lambda_s$, $\lambda_a \ll \mu$, $\lambda_s \perp \mu$ and these measure are positive, and $\lambda_a \perp \lambda_s$.

(ii) Prove that there is a Unique $b \in L'(\mu)$ such that $\lambda_a(E) = \int_E b d\mu \; (E \in m)$.

(8+7)

(*Or*)

(b) Prove that to each bounded linear functional Φ on $C_0(X)$, Where X is a locally compact Housdorff space, there corresponds a Unique complex regular Borel measure μ such that $\Phi(f) = \int_E f d\mu \; (F \in C_0(X))$. Also prove that if Φ and μ are related as in the above equation then $\|\Phi\| = |\mu|(X)$.

5. (a) Let (X, τ_1) and (Y, τ_2) are measurable spaces.

(i) If $E \in \tau_1 \times \tau_2$, then prove that $E_x \in \tau_2$ and $E^y \in \tau_1$ for every $x \in X$ and $y \in Y$.

(ii) Prove that $\tau_1 \times \tau_2$ is the smallest monotone class which contains all elementary sets.

(7 + 8)

(*Or*)

(b) State and prove Fubini Theorem.

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RW-6772 **571101**

M.Phil. DEGREE EXAMINATION APRIL 2011

Mathematics

COMMUTATIVE ALGEBRA

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

1. (i) Prove that an R-module P is projective, if and only if, for any surjective homomorphism $g : M \to M''$, the induced homomorphism $g^* : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, M'')$ is surjective.

   (ii) Let A be a faithfully flat R-algebra and M an R-module. If $A \otimes_R M$ is finitely generated and projective over A prove that M is finitely generated and projective over R.

(8 + 7)

(*Or*)

(b) Define Tensor product of two R-modules. Show that the Tensor product of two R-modules exists and it is unique up to isomorphism.

2. (a) (i) Define a local property. Is flatness a local property ? Justify your answer.

(ii) Show that an element $a \in J(R)$ if, and only if $1 + ab$ is a unit for all $b \in R$

(8 + 7)

(*Or*)

(b) (i) State and prove Nakayama's lemma.

(8 + 7)

(ii) Let S be a multiplicatively closed set and $f : R \to R_S$ be the natural map defined by $f(a) = \left(a/1\right)$. Then prove the following

(i) Every ideal of R_S is an extended ideal.

(ii) The prime ideals of R_S are in one-one correspondence with the prime ideals of R not intersecting S.

3. (a) (i) Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of R-modules. Prove that M is Noetherian if, and only if, both M' and M'' are Noetherian.

(ii) State and prove Hilbert's basis theorem.

(7 + 8)

(*Or*)

(b) (i) State and prove the structure theorem for Artinian rings.

(ii) Prove that an R-module M has a composition series if, and only if, it is both Noetherian and Artinian.

(9 + 6)

4. (a) (i) Show that an element $\alpha \in S$ is integral over R if and only if $R[x]$ is a finitely generated R-module, where R is a commutative ring with unity and S is an extension ring of R.

(ii) State and prove Going Down theorem.

(7 + 8)

(*Or*)

(b) State and prove Noether's Normalization theorem. Deduce Hilbert's Nullstellensatz theorem.

5. (a) (i) Let R be a subring of a field K and $h : R \to \Omega$ a nontrivial ring homomorphism of R into an algebraically closed field Ω. Prove that h has a maximal extension $\bar{h} : V \to \Omega$ where V is a valuation ring of K.

(ii) Let R be a subring of a field K. Prove that the integral closure $\bar{R}$ of R in K is the intersection of all valuation rings V of K containing R.

(8 + 7)

(*Or*)

(b) (i) Let R be a local domain. Prove that every non-zero fractionary ideal of R is invertible if , and only if, R is a discrete valuation ring.

(ii) Let R be a domain. Prove that R is a Dedekind domain if, and only if, every non-zero fractionary ideal of R is invertible.

(7 + 8)

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RW-6773 **571102**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Mathematics

FUNCTIONAL ANALYSIS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

Answer **five** questions, choosing **either** (a) **or** (b).

(5 × 15 = 75)

1. (a) Let X be a topological vector space, prove the following :

(i) If $A \subset X$ then $\bar{A} = \cap (A + V)$, where V runs through all neighbourhood of 0.

(ii) If $A \subset X$ and $B \subset X$, then $\bar{A} + \bar{B} \subset \overline{A + B}$.

(iii) If Y is s subspace of X, so is $\bar{Y}$.

(iv) If C is a convex subset of X, so arc $\bar{C}$ and C^o.

(v) If E is a bounded subset of X, so is $\bar{E}$.

(*Or*)

(b) Show that a topological vector space X is normable if and only if its origin has a convex bounded neighbourhood.

2. (a) State and prove Banach-Steinhaus theorem.

(*Or*)

(b) State and prove open-mapping theorem.

3. (a) If X is a locally convex space and H is the convex hull of a totally bounded set $E \subset X$, then show that H is totally bounded.

(*Or*)

(b) State and prove Krein-Milman theorem.

4. (a) Suppose U and V are the open unit balls in the Banach spaces X and Y respectively. Suppose $T \in \mathcal{B}(X, Y)$ and $e > 0$, prove that :

(i) If the closure of T (U) contains eV, then $T(U) \supset cV$.

(ii) If $c\| y^* \| \leq \| T^* y^* \|$ for every $y^* \in Y^*$, then $T(U) \supset cV$.

(*Or*)

(b) If X is a Banach space $T \in \mathcal{B}(X)$, T is compact and $\lambda \neq 0$, then show that $T - \lambda I$ has closed range.

5. (a) Let $[Z_o, \ldots, Z_n], \{W_o, \ldots, W_n\}$ be pick-Nevanlinna data. Let E be the set of all holomorphic functions in U such that $|f| < 1$ and $f(Zi) = Wi$, $0 \le i \le n$. If E is not empty, then show that E contains a finite Blaschke product.

(*Or*)

(b) Let G be a compact group, suppose $f \in C(G)$ and define $H_L(f)$ to be the convex hull of the set of all left translates of f. Then pProve that

(i) f is uniformly continuous and

(ii) $H_L(f)$ a totally bounded subset of C (G).

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RW-6774 **661104**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Commerce

RESEARCH FOR BUSINESS DECISIONS

(CBCS–2009 onwards)

Time : 3 Hours Maximum : 75 Marks

Section A (6 × 3 = 18)

Answer **all** questions.

All questions carry equal marks.

1. Who are entrepreneurs ?

2. Define brand equity.

3. What is Market Segmentation ?

4. What is brain storming ?

5. Explain Organisational Culture.

6. What is risk return trade off ?

Section B (4 × 6 = 24)

Answer any **four** questions.

All questions carry equal marks.

7. Firm L and firm U are in same risk class and are identical in every respect except that firm L is levered and firm 'U' Unlevered. Firm L has 6 percent Rs. 400,000 debentures outstanding. Both firms earn 12 percent before interest and taxes on their total assets which amounts to Rs. 800,000. Assume a corporate tax of 50 per cent and a pure equity of capitalisation rate of 10 percent compute total value of the firm under

   (a) NI approach.

   (b) NOI approach.

8. From the following data for project M calculate expected monetary value and standard deviation assuming a normal distribution. Calculate probability that the net present value will be zero or less, that it will be greater

than Rs. 20,000 and less than Rs. 2,000. Assume further that the cost of project in period ‘ O’ is Rs. 20,000 and its life is only one year

Net cash in flows in period I	Probability
Rs. 40,000	0.25
Rs. 48,000	0.50
Rs. 24,000	0.25

9. Discuss various functional areas of research.

10. Discuss demographic factors affecting consumer behaviour.

11. Discuss on the job training methods.

12. Discuss role of mutualfunds in mobilisation of savings from small investors.

Section C (3 × 11 = 33)

Answer **all** questions.

All questions carry equal marks.

13. (a) Discuss various factors affecting working capital management. Also explain various techniques of working capital management.

(*Or*)

(b) Underline the need for environmental research. Explain the trends in environmental research.

14. (a) Discuss the research on management of challenges on organisational change.

(*Or*)

(b) Describe the major traits of leadership. Draft a research design to study the relationship between the traits and leadership style.

15. (a) Explain various sources of recruitment along with their merits and demerits.

(*Or*)

(b) Describe the pricing strategies for a product based on the various stages in its life cycle.

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RW-6775 **581203**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Physics

THIN FILM PHYSICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

1. (a) Explain different sputtering method for film preparation.

(*Or*)

(b) Describe the different methods for film preparation.

2. (a) Describe the microbalance method to prepare thin films.

(*Or*)

(b) Discuss the different methods involved in surface characterization of thin films.

3. (a) Explain the effect of thickness on the conduction of metallic thin films.

(*Or*)

(b) Write about the different mechanisms of dielectric break down in thin films.

4. (a) Explain the operation of thin film photovoltaic solar cells.

(*Or*)

(b) Discuss the operation of PEC.

5. (a) Explain the different techniques involved in thickness measurement in thin films.

(*Or*)

(b) Explain the operation of a thermal evaporation apparatus with a neat block diagram.

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RW-6776 **581204**

M.Phil. DEGREE EXAMINATION APRIL 2011

Physics

SOLID STATE IONICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

Each question carries 15 marks.

1. (a) (i) Differentiate electronic and ionic conductors.

(ii) Explain the various structural factors responsible for ionic conduction.

(*Or*)

(b) (i) List the uses of super ionic materials.

(ii) Discuss the crystalline cationic and anionic conductors with examples.

2. (a) Explain the principle and application of X-ray diffraction technique in charcterising the super ionic materials.

(*Or*)

(b) Explain the various spectroscopic techniques involved in super ionic characterization. Discuss the merits and demerits of each technique.

3. (a) Describe the different type of computer modelling of superionics.

(*Or*)

(b) (i) Mention the advantages of solid state batteries.

(ii) Explain the phenomenon of mass transfer and thermodynamics in solid state batteries.

4. (a) Explain the double layer and other polarization effect in solid state interface.

(*Or*)

(b) (i) What are microionic materials ?

(ii) Explain the principle and working of micro ionic technology.

5. (a) Give the principle, working and applications of oxide fuel cells.

(*Or*)

(b) (i) Mention the applications of the sensing devices.

(ii) Explain the principle and working of electro chemical devices.

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RW-6777 **581205**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Physics

CRYSTAL GROWTH

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** the questions.
Each question carries 15 marks.

1. (a) Write short notes on : (i) Importance of crystal growth ; (ii) Crystal planes and ; (iii) Bragg's law.

(*Or*)

(b) Write briefly about the following : (i) Classification of crystals and (ii) Grain boundary and dislocations.

2. (a) Compare the slow cooling and slow evaporation methods for the growth of crystals.

(*Or*)

(b) Discuss the following : (i) Properties of gels ; (ii) Synerisis and (iii) Gel growth.

3. (a) Discuss the Czochralski technique for crystal growth, with suitable diagrams.

(*Or*)

(b) Write an essay on Kyrupoulos technique.

4. (a) Write in detail about the principles of flux growth, choice of flux and the technique of flux growth of crystals.

(*Or*)

(b) Write short notes on the following : (i) Bridgeman technique and (ii) High pressure crystal growth.

5. (a) Write short notes on : (i) Molecular beam method and (ii) Vapour phase crystallization in a closed system.

(*Or*)

(b) Discuss the chemical vapour transport method in C.V.D.

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RW-6777 **581205**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Physics

CRYSTAL GROWTH

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** the questions.
Each question carries 15 marks.

1. (a) Write short notes on : (i) Importance of crystal growth ; (ii) Crystal planes and ; (iii) Bragg's law.

(*Or*)

(b) Write briefly about the following : (i) Classification of crystals and (ii) Grain boundary and dislocations.

2. (a) Compare the slow cooling and slow evaporation methods for the growth of crystals.

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(b) Discuss the following : (i) Properties of gels ; (ii) Synerisis and (iii) Gel growth.

3. (a) Discuss the Czochralski technique for crystal growth, with suitable diagrams.

(*Or*)

(b) Write an essay on Kyrupoulos technique.

4. (a) Write in detail about the principles of flux growth, choice of flux and the technique of flux growth of crystals.

(*Or*)

(b) Write short notes on the following : (i) Bridgeman technique and (ii) High pressure crystal growth.

5. (a) Write short notes on : (i) Molecular beam method and (ii) Vapour phase crystallization in a closed system.

(*Or*)

(b) Discuss the chemical vapour transport method in C.V.D.

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RW-6778 **596102**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Chemistry

AREA OF SPECIALIZATION IN CHEMISTRY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

1. (a) (i) Write the basic terminology required for battery. (5)

(ii) Explain the principle and mechanism in primary system of battery. (10)

(*Or*)

(b) Elaborate the construction and working of Ni-Cd Battery.

2. (a) Give a detailed account on the Electrochemical method of measurement of corrosion rate by AC method.

(*Or*)

(b) Explain the process of Electroless plating of Nickel and Copper.

3. (a) Elaborate the Chemical constitution of colour suitable for Dyes.

(*Or*)

(b) Describe the preparation of Indigoid dye, Sulphur dye and Disperse dye. (3 × 5)

4. (a) (i) Explain with reaction, how β-d-glucose is converted into cellulose polymer ? (7)

(ii) Give the categories of amino acids in protein polymer. (3)

(iii) Write the differences between amino acids in Fibroin polymer and amino acids in Keratin polymer. (5)

(*Or*)

(b) Explain the mechanism of dyeing of

(i) Reactive dye, and.

(ii) Vat dye on cotton material. (7.5 + 7.5)

5. (a) (i) Describe the Kinetics of Cationic Polymerization. (8)

(ii) Comment on the application of liquid Crystalline polymers. (7)

(*Or*)

(b) (i) Discuss, how the T_g of a polymer is measured by DTA and TMA. (5 + 5)

(ii) Write on the factors affecting crystallinity of polymers. (5)

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RW-6779 **731102**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Education

ADVANCED EDUCATIONAL RESEARCH METHODOLOGY

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer the following questions in about 500–750 words each.

1. (a) What is scientific method ? Describe the various steps involved in scientific method highlighting their relevance to research process.

(*Or*)

(b) Due to lack of qualitative research, education, as a discipline, is still in the stage of infancy. Comment on this.

2. (a) What is research design ? Discuss its importance used in experimental hypothesis testing study.

(*Or*)

(b) Experiment research and expost facto research differ from each other. Comment with justification.

3. (a) How does a questionnaire differ from a schedule ? How will you establish validity of a questionnaire?

(*Or*)

(b) Define reliability and validity. How will you establish reliability and validity of a research tool by mentioning a tool.

4. (a) Explain various sampling techniques. Under what conditions the researcher uses random stratified sampling.

(*Or*)

(b) How do quantitative and qualitative approaches to educational research differ in data collection and use of findings.

5. (a) Distinguish between the following :

(i) Null hypothesis and Research Hypothesis.

(ii) Directional hypothesis and Non-directional hypothesis.

(iii) Type I error and Type II error.

(iv) Parametric test and non parametric test.

(v) Hypothesis and Assumptions.

(*Or*)

(b) What do you understand by historical criticism ? How does it help a researcher in establishing the authenticity of data.

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RW-6781 **921201**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Library and Information Science

AREA OF SPECIALIZATION : USER STUDIES

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

Each questions carries 15 marks.

1. (a) Describe the significant studies conducted on Information seeking behaviour in India.

(*Or*)

(b) Discuss the role of International Organization in conducting studies on Information seeking behaviour.

2. (a) Enumerate the taxonomy of Information needs and human needs.

(*Or*)

(b) Bring out the classification of Information needs.

3. (a) Explain the various pattern of user behaviour.

(*Or*)

(b) Discuss the limitations in Behavioural Research in Librarianship.

4. (a) Describe the methods of conducting user education.

(*Or*)

(b) Bring out the need and purpose for user Education.

5. (a) Define non-documentary sources and explain the various E-resources available for collecting information for a research.

(*Or*)

(b) Discuss the user interaction with documentary resources.

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RW-6788 **441201**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Women Studies

WOMEN AND UNORGANIZED SECTOR

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

Each questions carries 15 marks.

1. (a) Describe the role of women in work, family and decision making

(*Or*)

(b) Discuss the categorization of unorganised workers.

2. (a) Describe in the feminist perspective on struggles in the unorganised sector.

(*Or*)

(b) Explain the wage payment systems for the unorganised sector workers.

3. (a) What are the problems faced by women at workplace in unorganised sector ?

(*Or*)

(b) Define Gender and sex. Explain the sexual division of labour in unorganised sector.

4. (a) List out the hindrances while implementing policies for women in unorganised sector.

(*Or*)

(b) Bring out the programmes for women in unorganised sector.

5. (a) What are the problems faced by women worker in construction sector.

(*Or*)

(b) Discuss the economic and health problems of women worker in the stone quarries.

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RW-6789 **441202**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Women's Studies

WELFARE OF THE WEAKER SECTION

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** the questions.
Each question carries 15 marks.

1. (a) Who are nomads ? What are the Welfare Schemes provided by State Government for nomads and semi nomads communities in Tamilnadu.

(*Or*)

(b) Explain the Welfare schemes for the weaker sections provided by the present Government.

2. (a) Describe the status of differently abled women in India.

(*Or*)

(b) Define differently abled. What are the different types of differently abled ? Explain.

3. (a) Define mental retardation. Discuss about the mental retardation and Multiple Disability Act.

(*Or*)

(b) What do you mean by Autism, Cerebral palsy ? What are the facilities provided by the National Trust to them ?

4. (a) Describe the health problems of older women.

(*Or*)

(b) Explain the occupational and financial problem of old age women.

5. (a) Describe the mobile medical care services for the aged women.

(*Or*)

(b) List out the policies and programmes for old age people.

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RW-6790 **731201**

M.PHIL. DEGREE EXAMINATION, APRIL 2011

Education

MATHEMATICS EDUCATION

(CBCS—2010 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** the questions.
Each question carries 15 marks.

1. (a) Suggest the ways to motivate B.Ed. trainees in Mathematics through History of Mathematics.

(*Or*)

(b) Discuss the historical development of Mathematics in India.

2. (a) Discuss the Research trends in Mathematics Education.

(*Or*)

(b) Critically analyse the causes of under achievement in Mathematics

3. (a) Suggest the ways to bring about reforms in Mathematics curriculum at Higher secondary level.

(*Or*)

(b) Design a curriculum on 'Teaching of Mathematics' at B.Ed. level which could suit the student-teachers in the knowledge society.

4. (a) Discuss the theoretical bases of evaluation.

(*Or*)

(b) Prepare a diagnostic test in Mathematics at Higher secondary level.

5. (a) Is it imperative for a mathematics teacher to embrace modern techniques ? Justify.

(*Or*)

(b) Discuss the role of educational technology in Mathematics Education.

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RW-6793 **821201**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Physical Education

SPORTS PHYSIOTHERAPY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

1. (a) Describe basic massage technique and its effects.

(*Or*)

(b) Explain treatment of any two Sports injuries through yogic practices.

2. (a) Write in detail about Electrotherapy treatment.

(*Or*)

(b) Write in detail about Cryotherapy treatment.

3. (a) Briefly explain different types of strengthening exercise.

(*Or*)

(b) Discuss treatment for Lordosis and Kyphosis.

4. (a) Describe about prevention of athletic injuries.

(*Or*)

(b) Discuss about proprioceptive neuromuscular facilitation.

5. (a) Describe basic principles in application of functional bandage materials.

(*Or*)

(b) Describe different types of bandages.

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RW-6794 **821202**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Physical Education

SPORTS PHYSIOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

All questions carry equal marks.

1. (a) Explain the mechanism of sliding filament theory with the help of a diagram.

(*Or*)

(b) Describe the types of muscle fibre and their relation to sports performance.

2. (a) Explain the 'anaerobic' and 'aerobic' systems of energy production.

(*Or*)

(b) Describe the effects of exercise on muscular system.

3. (a) List down the physiological considerations of diet in relation to components, quantities and significance.

(*Or*)

(b) Distinguish between 'Sea Level' and 'high altitude' variations in the sports performance.

4. (a) Explain why the Doping substances are banned for the sportsmen.

(*Or*)

(b) Describe the effects of exercise on Carbohydrate, fat and protein metabolism.

5. (a) Distinguish between the 'aerobic' and anaerobic training schedules.

(*Or*)

(b) Explain the 'anaerobic threshold' and 'blood doping'.

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RW-6795 **821203**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Physical Education

SPORTS BIO–MECHANICS

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

Answer **all** questions.

Each questions carries 15 marks.

1. (a) Explain the nature and scope of Biomechanics in Physical Education.

(*Or*)

(b) Biomechanics aims at better performance– Justify.

2. (a) Describe various laws of motion by Newton and explain their application in sports activities.

(*Or*)

(b) Describe rotary and linear motion and explain the guiding principles derived from the application of mechanical principles motion and force.

3. (a) Describe buoyancy and fluid resistance and explain the guiding principles derived from the application of mechanical principles buoyance.

(*Or*)

(b) Explain the application of hydrodynamic principles on the swimming activity.

4. (a) Explain the mechanical principles wind resistance, form drug and surface drug and how it is applied in sports activities.

(*Or*)

(b) Explain the mechanical principles spin and gyration and describe the guiding principles derived from the application of the aerodynamics.

5. (a) Explain the application of Cinematographic video analysis in the field of physical education.

(*Or*)

(b) Describe the biomechanical analysis of Running and Jumping.

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RW-6796 **821204**

M.Phil. DEGREE EXAMINATION, APRIL 2011

Physical Education

SPORTS PSYCHOLOGY

(CBCS—2008 onwards)

Time : 3 Hours Maximum : 75 Marks

(5 × 15 = 75)

All questions carry equal marks.

1. (a) Define the term sports psychology ? Explain the influence of psychological factors in mortor learning and sports performance.

(*Or*)

(b) Define the term psychology and explain its meaning. How will you relate theory and practice in sports psychology ? Give suitable illustration.

2. (a) What is personality ? Explain the nature of personality. Explain the process of personality development.

(*Or*)

(b) How will you assess personality traits ? Describe the methods.

3. (a) Define 'stress' and Describe the causes of stress.

(*Or*)

(b) Describe the theories of stress and explain the factors of stress.

4. (a) What is motor learning ? Describe the learning principles and process of learning.

(*Or*)

(b) Define the term 'Anxiety'. Describe the relationship of Anxiety and sports performances.

5. (a) Explain the concepts of 'Attention' and 'perception'. How do they influence sports performance ?

(*Or*)

(b) Define the term 'cognitition'. Explain the cognitive behavioural intervention program.

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